IRT and Classical Item Analysis Report

Quiz 2 — summary for test authors
Based on the Stata log and supplied diagnostic graphs

Executive summary

• First attempts only: 52 students remained after 60 additional attempt records were removed (112 records before filtering, assuming one record per attempt).

• Average performance was 3.35 out of 5 correct (66.9%); scores ranged from 1 to 5, with 18 students (34.6%) scoring 4.

• Internal consistency was limited (Cronbach's alpha = 0.497), calculated on Items 2–5 because Item 1 was constant.

• Item 1 was correct for all 52 students and cannot discriminate among students in this sample. Item 2 was the most difficult (26.9% correct).

• Item 4 had the weakest item-rest correlation (0.199) and the shallowest 2PL curve; review its wording, scoring and construct alignment.

• The 2PL model did not materially improve fit over 1PL (LR chi-square(3) = 2.25, p = 0.5226); both AIC and BIC favor 1PL.

Data and analysis approach

The Stata workflow sorted records by student ID and attempt, retained the first attempt for each student, and created five dichotomous item scores (correct = 1, incorrect = 0). The analysis then calculated descriptive statistics, item-rest correlations, Cronbach's alpha, a common-discrimination 1PL model, and an item-specific-discrimination 2PL model. The log indicates that a response was coded correct only when its earned points equaled 15; the original quiz questions and scoring key were not provided for independent validation.

The total score is the sum of the five binary item indicators. The IRT models exclude Item 1, because all students answered it correctly; the IRT plots therefore describe four modeled items (Items 2–5).

Overall test performance

Statistic	Value	Interpretation
Students	52	Small classroom sample; item estimates are exploratory
Mean total score	3.346 / 5 (66.9%)	Moderately accessible quiz
Standard deviation	1.153	Some spread, but only five possible positive total scores observed
Observed range	1–5	No zero scores; 8 perfect scores
Cronbach's alpha	0.497	Limited internal consistency; Item 1 omitted as constant

Score distribution

The modal score was 4 correct (18 of 52 students, 34.6%). Half the students scored at most 3, and half scored 4 or 5. The frequencies were: score 1 = 4; 2 = 8; 3 = 14; 4 = 18; 5 = 8.

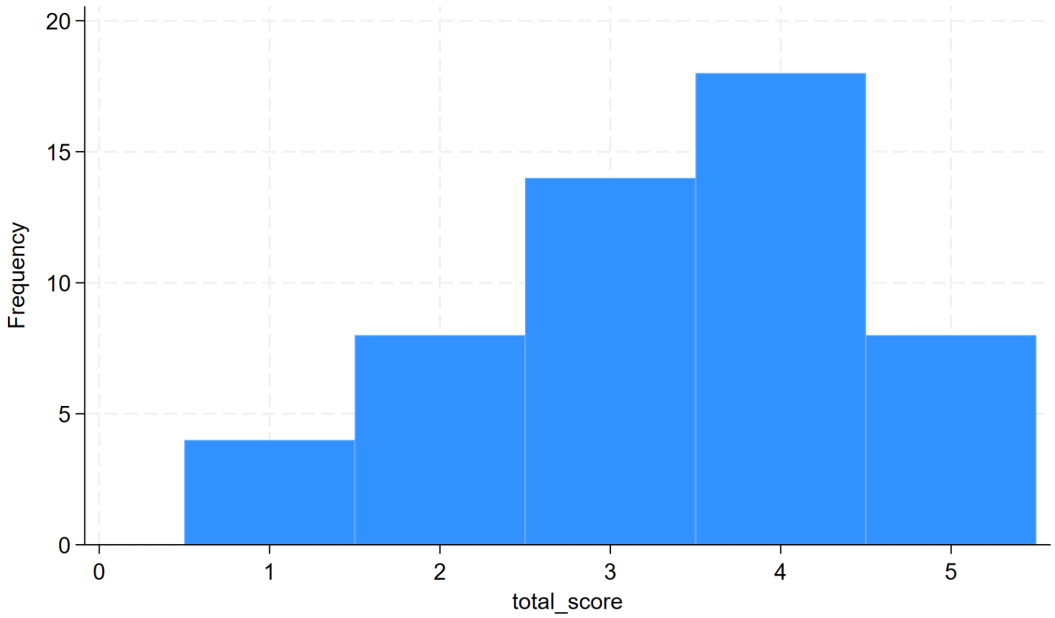


Figure 1. Observed distribution of total scores (five binary items).

Item-level findings

Item	Correct	Item-rest r	1PL difficulty	2PL discrimination	Author-facing interpretation
1	100.0%	Not defined	Not estimated	Not estimated	All students correct; retain only if intended as a warm-up or prerequisite check.
2	26.9%	0.274	1.033	1.271	Hardest; inspect difficulty, coverage, key and distractors.
3	69.2%	0.347	-0.839	1.695	Moderately easy; useful positive item-rest association.
4	59.6%	0.199	-0.404	0.587	Weakest item-rest relationship and shallowest 2PL curve; review first.
5	78.8%	0.362	-1.349	2.336	Easiest modeled item; comparatively strong item-rest relationship.

Difficulty is measured on the latent-ability scale: more positive values indicate harder items. Discrimination describes how sharply the probability of a correct answer increases as ability increases. These 2PL estimates have substantial standard errors, so differences in slopes should not be overinterpreted.

Item characteristic curves — 2PL

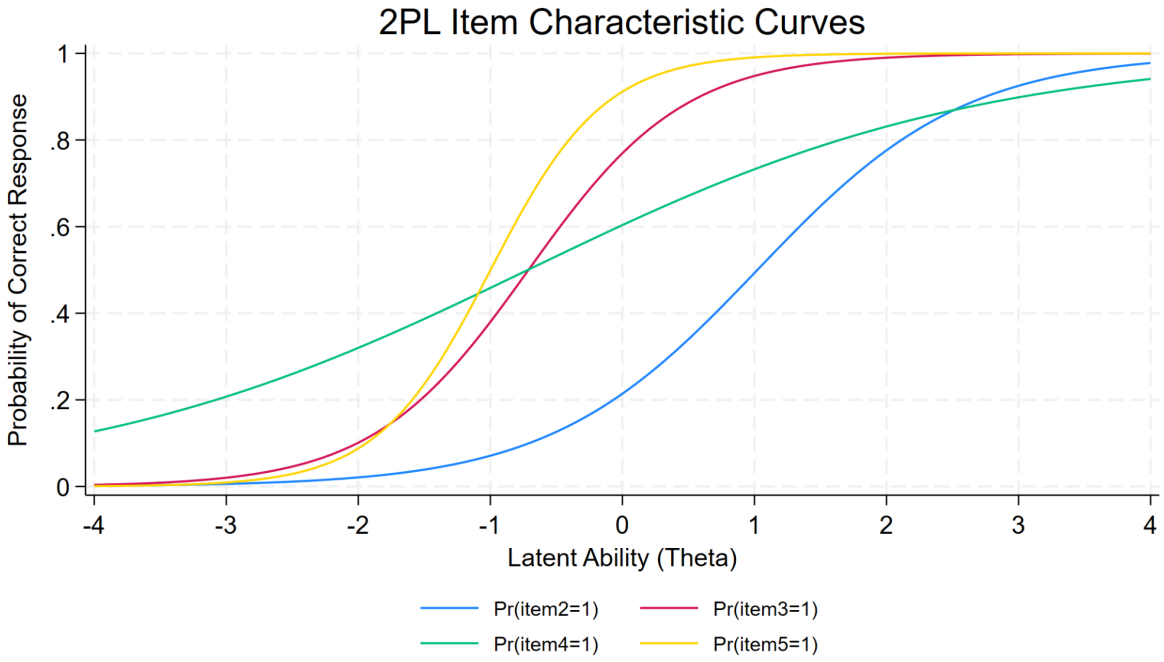


Figure 2. Modeled probabilities of correct responses across latent ability for Items 2–5.

Item 2 (blue) is shifted farthest right and is hardest. Item 5 (yellow) is shifted left and rises steeply; Item 3 (pink) also rises steeply. Item 4 (green) has a much flatter curve, indicating weaker estimated discrimination over a broad ability range. Curve crossings are possible in the 2PL model because items have different discrimination parameters. At theta = 0, Item 2 has a relatively low predicted success probability while Items 3–5 are substantially higher.

Test characteristic curve

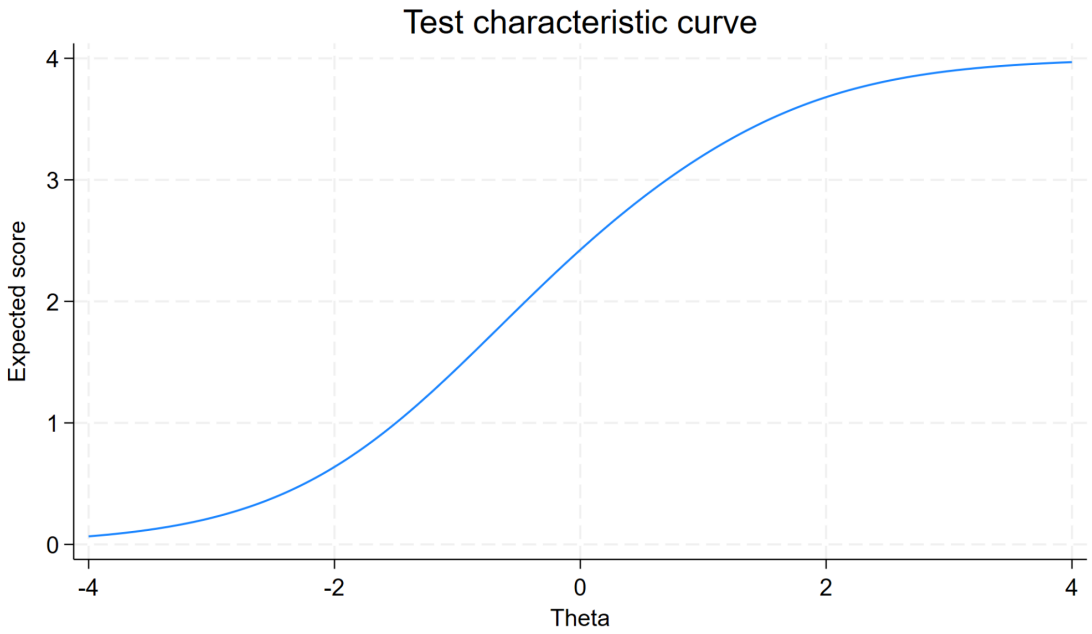


Figure 3. Expected number correct among the four IRT-modeled items as a function of theta.

The expected score increases monotonically from near zero at low ability toward four at high ability. At theta = 0, the expected modeled score is about 2.4 out of 4. This is not the expected score out of five, because the constant Item 1 was excluded from the IRT model. If Item 1 is treated as invariably correct, one would add one to obtain a corresponding five-item expected score.

Test information function

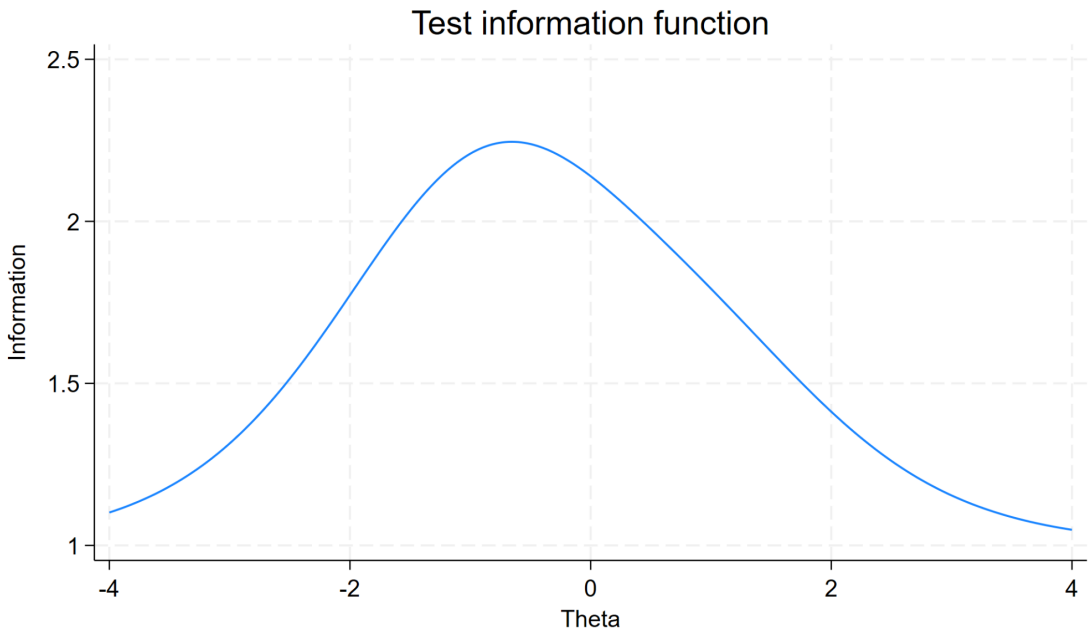


Figure 4. Test information curve from the 1PL model (generated before the 2PL model was fitted).

Information peaks near theta = -0.7 (approximately 2.25 on the graph), suggesting the modeled questions measure students somewhat below average latent ability most precisely. Information falls toward both extremes, so individual ability estimates are less precise for very low- or high-ability students. The supplied TIF and TCC were generated after the 1PL fit, before the 2PL fit; they should be interpreted as 1PL diagnostics, not as 2PL diagnostics.

Reliability and item relationships

Cronbach's alpha was 0.497 for Items 2–5. Corrected item-rest correlations were 0.274 (Item 2), 0.347 (Item 3), 0.199 (Item 4), and 0.362 (Item 5). Alpha if Item 4 is deleted rises slightly to 0.517, whereas deleting Item 3 or 5 lowers alpha to 0.372 or 0.368. These changes are small and based on only 52 students, so they are diagnostic rather than definitive.

Pairwise correlations among Items 2–5 were all positive but modest. The largest was Item 3 with Item 5 (r = 0.369). The smallest was Item 3 with Item 4 (r = 0.131). These patterns support reviewing Item 4 while also recognizing that a short quiz covering multiple distinct concepts need not have high internal consistency.

Comparison of 1PL and 2PL models

Measure	1PL	2PL
Log likelihood	-119.173	-118.049
Parameters	5	8
AIC	248.345	252.098
BIC	258.102	267.708
Likelihood-ratio test	—	chi-square(3) = 2.25; p = 0.5226

The 2PL fit has a slightly higher log likelihood, as expected from its greater flexibility, but the improvement is not statistically compelling. Lower AIC and BIC favor the more parsimonious 1PL model for this dataset. The 2PL plot is useful as an exploratory diagnostic, not strong evidence that the items truly differ in discrimination.

Recommended actions for quiz authors

1. Review Item 4 first: inspect the wording, scoring key, distractors and alignment with the intended mathematical skill. It has the weakest item-rest relationship and flattest fitted 2PL curve.

2. Check Item 2 for intended difficulty: only 14 of 52 students answered correctly. Determine whether it assesses an appropriately challenging concept or contains avoidable obstacles.

3. Decide the role of Item 1: because everyone answered it correctly, it may work as a confidence-building warm-up, but it adds no observed discrimination to this sample.

4. Keep Items 3 and 5 provisionally: they have the stronger item-rest correlations; review their content as well before treating them as anchor items.

5. Add several items across difficulty levels and collect more first-attempt responses before using theta to evaluate individual progress or compare semesters.

6. After revisions, repeat the same first-attempt analysis and inspect the item-rest correlations, reliability, 1PL/2PL comparison and information curve.

Cautions and assumptions

The results depend on the binary scoring threshold used in the Stata code. Stata’s rowtotal() can effectively count missing responses as zero; distinguish skipped answers from missing technical data if that matters. The sample is small for fitting eight parameters in a 2PL model, and individual discrimination standard errors are large. Ability is a relative latent scale within this quiz and cannot be directly compared with another quiz without common-item calibration or a linking design.

The source log contains a few stale code comments referring to six items or Item 1 graphs; the executed commands and numerical results consistently show five scored items, with Items 2–5 fitted in IRT. This report follows the executed commands.

Appendix: key model outputs

1PL: N = 52; log likelihood = -119.17265; common discrimination = 1.25164 (SE 0.33899); item difficulties: Item 2 = 1.03275, Item 3 = -0.83917, Item 4 = -0.40374, Item 5 = -1.34897.

2PL: N = 52; log likelihood = -118.04888; discriminations: Item 2 = 1.27124, Item 3 = 1.69510, Item 4 = 0.58681, Item 5 = 2.33587; difficulties: Item 2 = 1.02240, Item 3 = -0.71089, Item 4 = -0.71588, Item 5 = -0.99844.